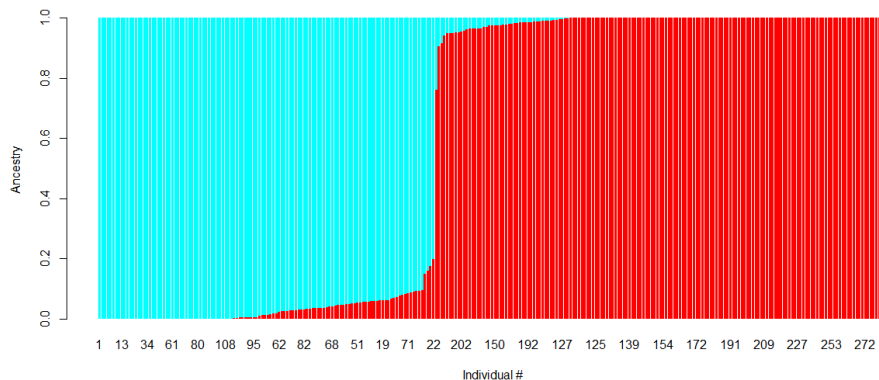


Admixture

Part I: Samples of known ancestry

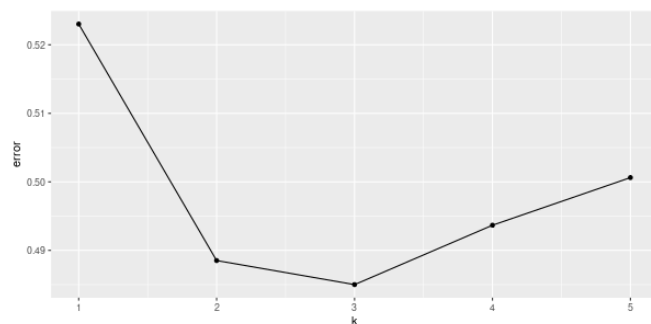


What does this plot tell us about the ancestry of the samples?

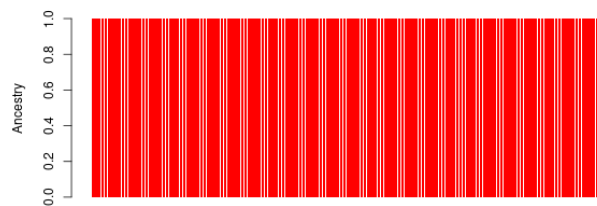
It shows us that there is a decent range of mixtures in the population, but most people are mostly African or European.

Part II: Samples of unknown ancestry

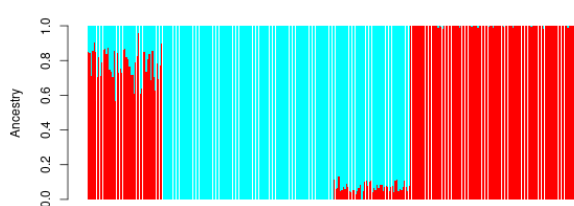
What do the plots tell us about the populations?



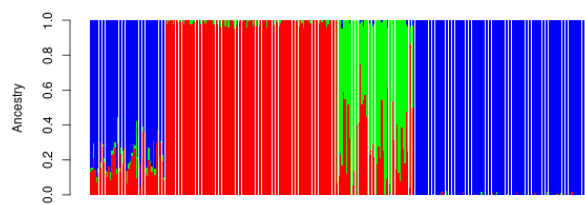
From the first plot we can see that there is mostly likely 2 or 3 populations amongst our data, because these values of k minimize our error. From the plots below we can see how the clustering algorithm groups the different individuals into different groups (depending on what k equals) as seen from the color.



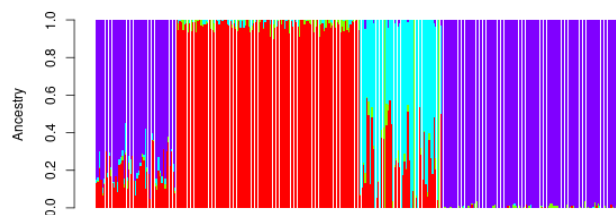
Individual #



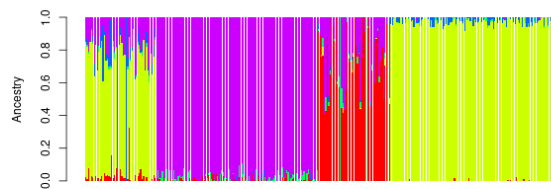
Individual #



Individual #



Individual #



Individual #

Power Calculations

1. Please define the following in words
 - a. What is type I error? (*two sentences*)
It is a false positive. This means that the null hypothesis has been rejected when it is true.
 - b. What is type II error? (*two sentences*)
It is a false negative. This means that we failed to reject the null hypothesis when it should have been rejected.
2. Using Hardy Weinberg Equilibrium, calculate the expected genetic exposure frequency, $P(G=1)$, for a SNP with a risk allele frequency of $p=0.1$ and $p=0.5$ under the following inheritance models. Remember: HWE: $p^2 + 2pq + q^2 = 1$

- a. Dominant inheritance (Homo- or heterozygous):

$p = 0.1$:

$$p^2 + 2pq = (0.1)^2 + 2(0.1)(0.9) = 0.19$$

$p = 0.5$:

$$p^2 + 2pq = (0.5)^2 + 2(0.5)(0.5) = 0.75$$

- b. Recessive inheritance (Homozygous only):

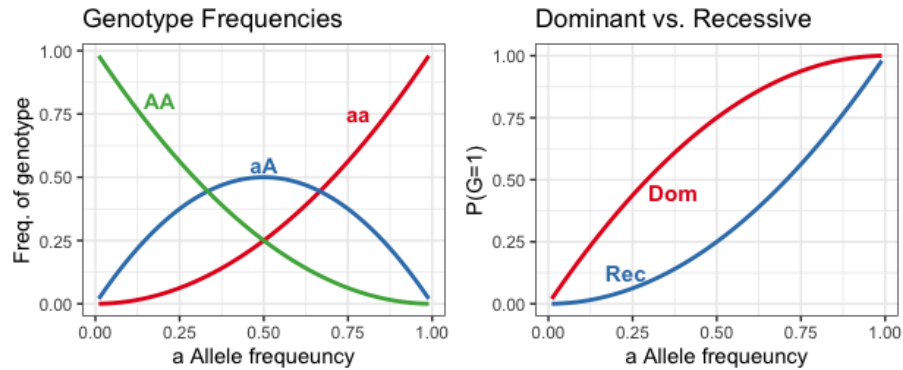
$p = 0.1$:

$$p^2 = (0.1)^2 = 0.01$$

$p = 0.5$:

$$p^2 = (0.5)^2 = 0.25$$

- c. What do you think are the possible implications of dominant vs. recessive inheritance with regard to power and sample size in genetic epidemiology studies? The figure below may help.



In the case of dominant inheritance you would need a higher allele frequency compared to the recessive inheritance to get the same amount of power. Thus you would need to have a higher sample size to get the same power when you have recessive inheritance, because your representative population are only the homozygous recessive, while for the dominant inheritance your representative population includes both the heterozygous and homozygous dominant individuals.

3. What is needed to determine the sample size for a case-control study focused on tobacco smoke exposure? Which parameters must be specified? (*three sentences*)

To determine the sample size for a case-control study focused on tobacco smoke exposure, we need to specify the power. This is because a larger power can cause an increase in sample size. Additionally, the type 1 error rate, OR/effect size, and variance in the outcome population are needed to determine the sample size.

4. Perform a power assessment for an association study of common SNPs within the VAKSHUN gene. Using GxE power - https://camadams.shinyapps.io/power_gxe/ - calculate power for a case-control study design.

Your disease outcome (yes/no) is rheumatoid arthritis case-status with a population prevalence of 2%. Calculate power for dominant and recessive models, *assuming* the following:

- Sample size is 2,000 cases with 1 control per case
- an allele frequency range of 0.1 - 0.5 in controls/population
- Genetic effect size range of OR=1.1 - 1.5
- type 1 error rate (alpha), two-sided of 0.05

Copy/paste screenshot of results below and describe using 2-3 sentences.

Power

	Pg									
	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5	
ORg										
1.1	0.223	0.275	0.307	0.322	0.325	0.316	0.300	0.278	0.251	
1.15	0.425	0.519	0.571	0.595	0.597	0.583	0.555	0.516	0.467	
1.2	0.641	0.748	0.799	0.819	0.820	0.806	0.778	0.738	0.683	
1.25	0.815	0.898	0.929	0.940	0.940	0.931	0.914	0.885	0.842	
1.3	0.922	0.968	0.981	0.985	0.985	0.981	0.973	0.959	0.933	
1.35	0.972	0.992	0.996	0.997	0.997	0.996	0.993	0.988	0.976	
1.4	0.992	0.998	0.999	1.000	1.000	0.999	0.999	0.997	0.992	
1.45	0.998	1.000	1.000	1.000	1.000	1.000	1.000	0.999	0.998	
1.5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.999	

N=2000 cases, 1 controls per case
Inheritance: Dominant
alpha=0.05
of independent tests=1

In the dominant model, under small effect sizes, there is low power across the board regardless of frequency of the gene. As you start to increase the OR you start to increase the power.

Power

	Pg								
	0.1	0.15	0.2	0.25	0.3	0.35	0.4	0.45	0.5
ORg									
1.1	0.049	0.066	0.087	0.112	0.140	0.170	0.201	0.232	0.261
1.15	0.066	0.100	0.143	0.195	0.255	0.318	0.381	0.441	0.494
1.2	0.087	0.143	0.216	0.305	0.401	0.497	0.585	0.661	0.722
1.25	0.111	0.195	0.305	0.432	0.559	0.672	0.764	0.832	0.879
1.3	0.139	0.256	0.405	0.563	0.704	0.812	0.886	0.932	0.959
1.35	0.171	0.324	0.509	0.684	0.819	0.906	0.954	0.978	0.989
1.4	0.207	0.397	0.609	0.786	0.900	0.958	0.984	0.994	0.998
1.45	0.245	0.471	0.701	0.864	0.949	0.984	0.995	0.999	1.000
1.5	0.286	0.545	0.779	0.919	0.977	0.994	0.999	1.000	1.000
N=2000 cases, 1 controls per case									
Inheritance: Recessive									
alpha=0.05									
# of independent tests=1									

With the recessive model, across all values you see lower power, especially when the allele is rarer. At lower allele frequencies, we see a drop in power when looking at recessive inheritance compared to the dominant inheritance model.

- Sample size assessment for association study of a single environmental exposure. You are interested in studying exposure to Epstein-Barr virus (EBV) and its relationship to multiple sclerosis (MS) susceptibility. Specifically, you want to collect self-report data on the diagnosis of infectious mononucleosis before age 18 in a group of cases and controls. Assuming the following:
 - Assume 1 control per case
 - Control (population) prevalence for infectious mononucleosis before age 18 (binary; yes or no) is 8%

- $OR_e = 1.5, 2.0, 3.0, \text{ or } 4.0$
- Power = 80%
- Type 1 error rate (α) = 0.05

Calculate the sample size needed detect effect sizes described above given the input parameters. *Copy/paste screenshot of results below and describe using 2-3 sentences (include sample size results in your description).*

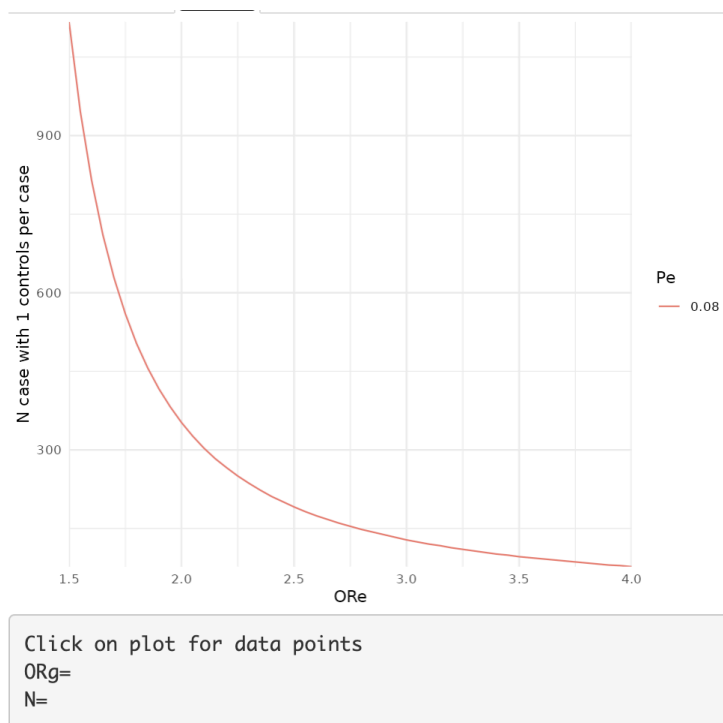
OR - Sample Size

1.5 - 1117

2 - 352

3 - 128

4 - 77



As the odds ratio increases, the required sample size decreases from 1117 to 77. The biggest drop is when we go from 1.5 - 2, and then the graph decreases exponentially. Once we get to the $OR = 2$, the required sample size decreases less exponentially, and at $OR = 3$, the sample size further decreases until we get to the odds ratio around 4.

6. Sample size assessment for detecting GxE interaction and association with MS. Assuming the following:

- A SNP in the *CR2* gene has (allele) frequency of 25% in the population and is believed to increase susceptibility to EBV infection.
- Dominant inheritance
- Infectious mononucleosis before age 18 at 12% prevalence

- Genetic Effect: $OR_g = 1.2$
- Environmental Effect: $OR_e = 1.2$
- GxE interaction effect: $OR_{gxe} = 1.5-2.0$
- type 1 error rate (α)=0.05
- 80% power

What sample size do you need when you assume 1 control for each case?

The sample size ranges from 1149 to 3095 depending on the OR_g (ranging from 1.5 to 2.0).

What sample size do you need when you assume 4 controls for each case?

The sample size ranges from 547 to 1632 depending on the OR_g (ranging from 1.5 to 2.0).

Consider a scenario where you want to test 1000 SNPs. How does this affect sample size?

The sample sizes increased dramatically.

How does the sample size required for GxE interaction compare to looking at genetic or environmental effects only? What are the practical implications for studying GxE?

Copy/paste screenshot of results below and describe using 2-3 sentences (include sample size results in your description).

Sample size N

Pg	
0.25	
ORg	
1.5	3095
1.55	2671
1.6	2341
1.65	2079
1.7	1867
1.75	1692
1.8	1547
1.85	1424
1.9	1319
1.95	1228
2	1149

N is the number of cases required for input parameters
M = 2000 control(s) per case
Pe=0.12
Inheritance: Dominant
alpha=0.05; beta=0.80
of independent tests=1

Sample size N

Pg	
0.25	
ORg	
1.5	1632
1.55	1392
1.6	1207
1.65	1060
1.7	942
1.75	845
1.8	764
1.85	697
1.9	639
1.95	590
2	547

N is the number of cases required for input parameters
M = 2000 control(s) per case
Pe=0.12
Inheritance: Dominant
alpha=0.05; beta=0.80
of independent tests=1

The screenshot shows the same table as the previous one, but with sample size values that are approximately 10 times larger due to the increased number of independent tests (1000 instead of 1).

Pg	
0.25	
ORg	
1.5	4986
1.55	4253
1.6	3686
1.65	3238
1.7	2876
1.75	2581
1.8	2335
1.85	2128
1.9	1952
1.95	1802
2	1671

N is the number of cases required for input parameters
M = 1 control(s) per case
Pe=0.12
Inheritance: Dominant
alpha=0.05; beta=0.80
of independent tests=1000

Similar to what we saw earlier, when you have a larger effect size, you need a larger sample size to conduct your study. When we increase the controls per case to 4 then the sample size you need dramatically decreases for the given effect size. By increasing the number of tests, we would expect the sample size to increase, because you would need to be a lot more stringent (some sort of adjustment like Bonferroni). Furthermore, the practical implication of using an interaction term like GxE is that it would increase the sample size.